AMAR NATH CHAURASIYA

📍 Gorakhpur, Uttar Pradesh, India
📧 chaurasiyaamarnath96@gmail.com | 📱 +91-7379056759
🔗 LinkedIn: linkedin.com/in/your-profile
🔗 GitHub: github.com/chaurasiyaamarnath96-coder

PROFESSIONAL SUMMARY

Machine Learning and Computer Vision enthusiast pursuing a B.Tech in Computer Science & Engineering with hands-on experience in developing AI-powered applications using Deep Learning, Computer Vision, and Data Analysis techniques. Skilled in Python, Machine Learning, Deep Learning, SQL, OpenCV, and FastAPI. Experienced in building end-to-end AI solutions for industrial inspection, agriculture, and healthcare domains. Passionate about solving real-world problems through data-driven technologies.

EDUCATION

Maharaja Agrasen College of Engineering & Technology

Bachelor of Technology (B.Tech) in Computer Science & Engineering

Graduation: 2026
CGPA: 7.6

Government Polytechnic Gorakhpur , Uttar Pradesh

Diploma in Mechanical Engineering
2022 | Percentage: 79%

TECHNICAL SKILLS

Programming Languages

- Python
- SQL

Machine Learning & AI

- Machine Learning
- Deep Learning
- Computer Vision
- Neural Networks

- Classification Models
- Object Detection
- Generative AI and RAG Pipeline

Frameworks & Libraries

- OpenCV
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- FastAPI
- Streamlit

Tools & Technologies

- Git
- GitHub
- VS Code
- Jupyter Notebook
- YOLOv11n
- Docker
- Google Colab

Databases

- MySQL

EXPERIENCE

Machine Learning Intern | Unified Mentor Pvt Ltd

- Worked on Machine Learning and Data Analysis projects involving data preprocessing, feature engineering, model development, and evaluation.
- Developed predictive models using Python and ML libraries.
- Performed exploratory data analysis and created insightful visualizations.

- Collaborated on practical AI solutions and technical documentation.
- Applied industry-standard development workflows using Git and GitHub.

PROJECTS

1. RAG-Based Document Question Answering System

GitHub: [chaurasiyaamarnath96-coder/RAG-Based-Document-Question-Answering-System]

- Developed an end-to-end Retrieval-Augmented Generation (RAG) application for querying information from PDF and document collections.
- Implemented document ingestion, text chunking, embedding generation, vector indexing, and semantic retrieval pipelines.
- Integrated Large Language Models (LLMs) with vector search to generate context-aware responses.
- Built an interactive Streamlit interface for document upload and question answering.
- Improved answer accuracy by grounding LLM responses on retrieved document context, reducing hallucinations.
- Designed scalable retrieval workflows using embedding models and vector databases.

Tech Stack: Python, LangChain, FAISS/ChromaDB, Hugging Face, Streamlit, LLMs, RAG

2. Welding Defect Detection System

GitHub: github.com/chaurasiyaamarnath96-coder/welding_defect_detection_app

- Developed an AI-powered welding defect detection application for industrial quality inspection.
- Trained and deployed a YOLOv11n object detection model for automated defect identification.
- Built an interactive Streamlit interface for image upload and real-time prediction.
- Implemented image processing and visualization techniques to highlight detected defects.
- Improved manufacturing inspection efficiency by reducing manual quality assessment.

Tech Stack: Python, YOLOv11n, OpenCV, Streamlit, Deep Learning

---

3. Plant Disease Detection System

GitHub: github.com/chaurasiyaamarnath96-coder/plant-disease-capstone

- Developed a Computer Vision-based system to identify plant diseases from leaf images.
- Applied image preprocessing and augmentation techniques to improve model performance.

- Built predictive models for disease classification across multiple plant categories.
- Designed an easy-to-use interface to support early disease diagnosis.
- Enabled data-driven decision support for agricultural applications.

Tech Stack: Python, Pytorch, OpenCV, NumPy, Streamlit

---

4. Face Mask Detection System

GitHub: github.com/chaurasiyaamarnath96-coder/AI_mask_detction_system

- Built a real-time face mask detection system using Deep Learning and Computer Vision.
- Classified masked and unmasked faces with high prediction accuracy.
- Implemented image preprocessing pipelines for robust detection performance.
- Developed a user-friendly interface for monitoring and prediction.
- Demonstrated practical AI application for public health and safety monitoring.

Tech Stack: Python, OpenCV, Pytorch, Deep Learning, FastAPI

ACHIEVEMENTS

- Completed multiple end-to-end Machine Learning and Computer Vision projects.
- Developed real-world AI applications in healthcare, agriculture, and industrial inspection domains.
- Actively practicing Data Structures & Algorithms and problem-solving.
- Strong understanding of model training, evaluation, deployment, and Git-based development workflows.

SOFT SKILLS

- Problem Solving
- Analytical Thinking
- Team Collaboration
- Communication
- Time Management
- Continuous Learning

CERTIFICATIONS

- Machine Learning Internship Certificate – Unified Mentor
- Relevant certifications in Python, AI, Data Analysis, and Machine Learning – Advance Machine Learning Specialization by Coursera